MIL-D-195B
6 July 1981
SUPERSEDING
MIL-D-195A
31 May 1967

MILITARY SPECIFICATION

DRUMS: STEEL, CALCIUM CARBIDE

(100-POUND CAPACITY)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a calcium carbide shipping drum.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-S-92	- Screw, Machine, Slotted, Cross Recessed or Hexagon Head.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8110

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specification and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A109	- Steel, Carbon, Cold-Rolled Strip.
A659	- Steel, Carbon (0.16 Maximum to 0.25 Maximum Percent), Hot Rolled Sheet and Strip Commercial Quality.
D997	- Drop Test for Cylindrical Shipping Containers.
D998	- Penetration of Liquids into Submerged Loaded Shipping Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. The drum shall be as shown in figure 1 and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.2.1 Steel sheet. Sheet steel shall conform to ASTM A659, UNS G 10150 or G 10200.

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3.2.2 Steel strip. Steel strip shall conform to ANSI/ASTM A109, Temper No. 4, Finish No. 1.

3.2.3 Gasket material. Gasket material shall be of synthetic rubber.

3.2.4 Sealing compound. The sealing compound shall be of a commercially approved nonhardening type of material that will provide a waterproof seal when used as specified herein.

3.3 Design and construction. The design and construction of the drum shall be as shown in figure 1 and as specified herein.

3.3.1 Drum. The drum shall be constructed of steel as specified herein having a minimum thickness of 22 gauge (0.0299 inch).

3.3.1.1 Body seam. The longitudinal body seam shall be lap welded.

3.3.1.2 Rolling hoops. Two rolling hoops shall be provided in the body of the drum as shown in figure 1.

3.3.1.3 Chimes. The top and bottom heads shall be attached to the body by double seams which provide for five thicknesses of metal at the chime. The sealing compound (see 3.2.4) shall be applied between the seams. Excess seaming compound shall be removed from the inside and outside of the drum.

3.3.1.4 Cover. The top head of the drum shall be fitted with a cover as shown in figure 1.

3.3.2 Closure.

3.3.2.1 Gasket. A tubular or flowed-in gasket (see 3.2.3) shall be attached to the cover and shall remain in place when the cover is removed. The gasket material shall be of sufficient thickness and resiliency to provide a seal that will not leak when tested in accordance with 4.4.2.1 and 4.4.2.3.

3.3.2.2 Closure fasteners. The cover shall be attached to the top head of the drum by means of four 5/16-inch slotted hexagon head machine screws spaced equidisant in a 10 1/2-inch circle. The screws shall screw into four 5/8-inch closed end nuts that are securely attached to the top head of the drum. The screws shall conform to FF-S-92 type 1, style 10s.

3.4 Performance.

3.4.1 Leakage resistance. The drums shall not leak when subjected to an internal air pressure of 5 pounds per square inch (psi) (see 4.4.2.1).

3.4.2 Drop resistance. The drums shall not rupture and the cover shall not fail when tested as specified in 4.4.2.2.

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3.4.3 Liquid penetration resistance. The drums shall not permit liquid to penetrate into the interior when tested as specified in 4.4.2.3.

3.5 Identification marking. The drum shall be marked for identification in accordance with MIL-STD-130.

3.6 Treatment and painting. The exterior of the drum and cover shall be cleaned, treated, and painted in accordance with MIL-T-704, type C. Color and finish shall be as specified (see 6.2).

3.7 Workmanship. The interior of the drum shall be cleaned and free of all loose scale, rust, dirt, and other foreign matter. The bolts and nuts shall be free of rust and damaged threads. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to either the appearance or strength of the weldment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and testing shall be in accordance with MIL-STD-105, level S-2.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 1.5 percent defective for major defects and 2.5 percent defective for minor defects.

4.3.3 Tests.

4.3.3.1 Individual. Each drum shall be tested as specified in 4.4.2.1. Failure of the test shall be cause for rejection of the drum.

4.3.3.2 Samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.2 and 4.4.2.3. AQL shall be 1.5 percent defective.

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4.4 Inspection procedure.

4.4.1 Examination. Drums shall be examined for the following defects:

Major

101. Materials not as specified.
102. Dimensions not as specified.
103. Body not as specified.
104. Body seam not as specified.
105. Rolling hoops not as specified.
106. Chimes not as specified.
107. Cover not as specified.
108. Closure fasteners not as specified.
109. Gasket missing or does not remain in place when cover is removed.
110. Treatment and painting not as specified.
111. Workmanship not as specified.

Minor

201. Seaming compound visible inside or outside of the drum.
202. Marking not as specified.
203. Color not as specified.

4.4.2 Tests.

4.4.2.1 Leakage. Each drum shall be fitted with a test cover through which air can be induced. The drums shall be submerged in water so that the uppermost surface is beneath the water surface no less than one inch and no more than two inches. An internal air pressure of five psi shall be applied and any evidence of leakage shall constitute failure of the test.

4.4.2.2 Drop. The sample drums shall be filled to within 1/2-inch of the top with a mixture of sand and water weighing 100 pounds. The closure shall be securely fastened and the filled drum shall be tested in accordance with ASTM D997. Evidence of rupturing of the seams, destruction of the chimes, or separation of the cover that causes exposure or spilling of the contents shall constitute failure of this test.

4.4.2.3 Liquid penetration. Sample drums, after passing the drop test, shall be tested for liquid penetration in accordance with ASTM D998. Evidence of liquid penetration shall constitute failure of this test. Leakage shall be determined by weighing the sample drum before and after submersion and by visual examination of the contents after submersion.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112.	Closure of drums not as specified.
113.	Palletization not as specified when required.
114.	Marking illegible, incorrect or incomplete.

5. PACKAGING

5.1 Packing. Each drum shall have cover and gaskets in place and the bolts hand tightened. Unless otherwise specified herein, the drums shall be shipped loose. When specified (see 6.2), the drums shall be palletized for shipment. The drums shall be stacked three tiers high on a pallet in accordance with MIL-STD-147, load type III or IIIa.

5.2 Marking. (see 6.3).

5.2.1 Level A. Marking designated for level A shipment shall be in accordance with MIL-STD-129.

5.2.1 Commercial. Marking designated for commercial shipments shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The drums are specifically intended for use in packing, storing and shipping of calcium carbide in 100 pound lots.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Color and finish required (see 3.6).
(c) When palletization is required (see 5.1).
(d) Markings required for presentation/packing (see 5.2 and 6.3).

6.3 Preservation/packing markings. The nature of this item is such that preservation is not required and the packing requirements contained in Section 5 are the minimum requirements necessary. For the purpose of marking, degrees of preservation/packing shall be designated A/A or Commercial/Commerical as specified (see 6.2).

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6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

Custodians:
Army - ME
Navy - SA
Air Force - 69

Review activity:
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User activities:
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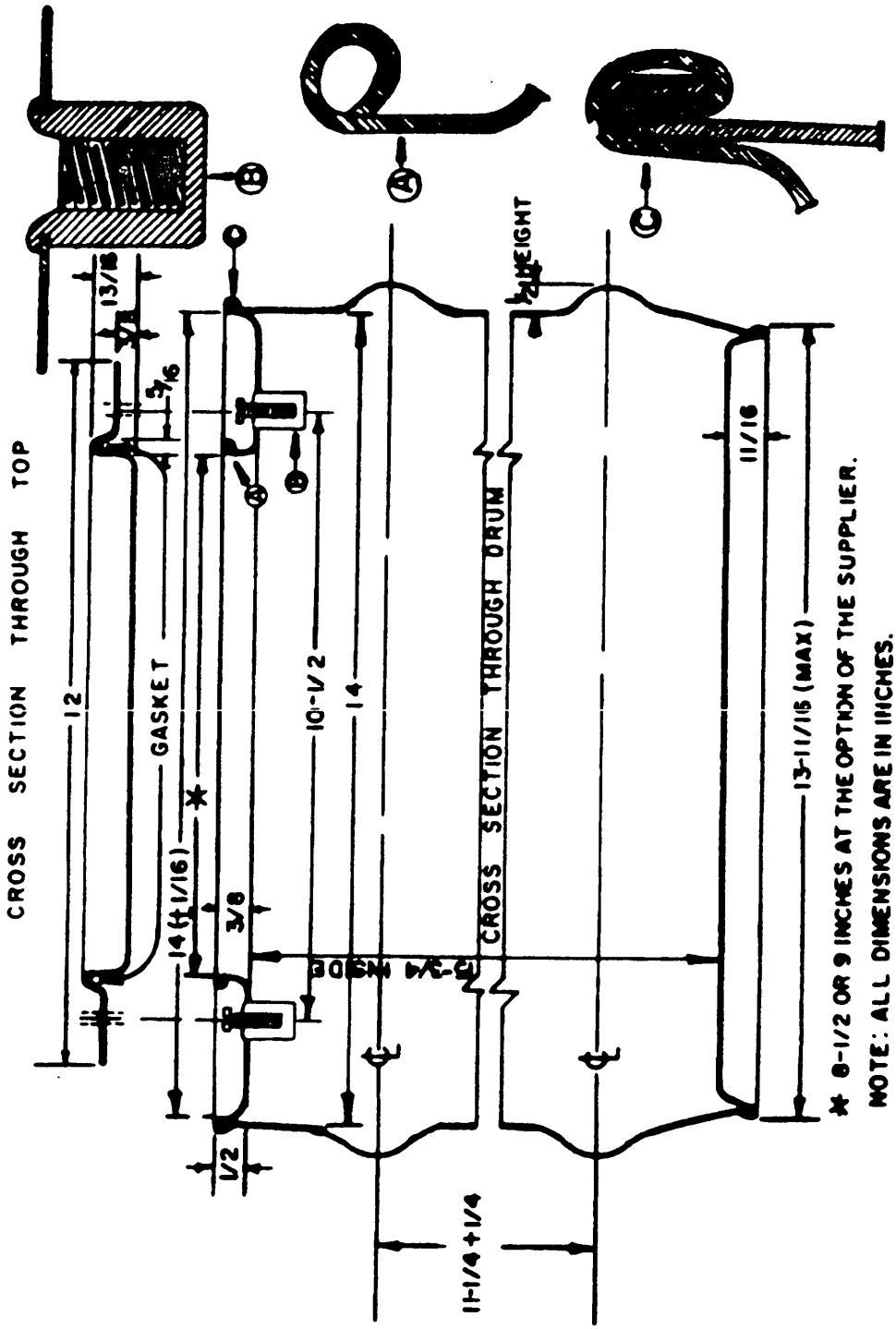


FIGURE 1. Steel, drum for 100 lbs calcium carbide.

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